← Back **Check your understanding: End of module 1**
Graded Assignment • 20 min

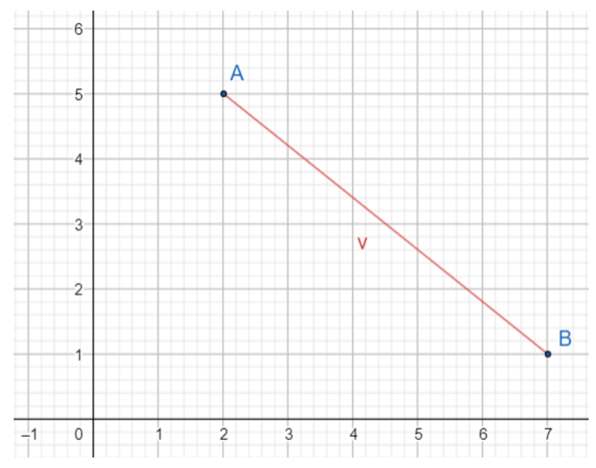
Due Jul 1, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Look at the vector v from A to B below. 1 / 1 point



Select all correct statements.

- [ ] $v = \overrightarrow{BA}$
- [x] $A = (2, 5)$

  Nice work

- [x] $v = \overrightarrow{AB}$

  Nice work

- [x] $v = \begin{pmatrix} +5 \\ -4 \end{pmatrix}$

  Nice work

- [ ] $v = \begin{pmatrix} +4 \\ -5 \end{pmatrix}$
- [x] $v = \begin{pmatrix} 7 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \end{pmatrix}$

  Nice work

- [x] $B = (7, 1)$

  Nice work

2. Compute the following dot product of vectors: 1 / 1 point

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

Select all correct options.

- [x] $= 11$

  Nice work

- [ ] $= 1 \times 3 \times 2 \times 4$
- [ ] None of the other options is correct.
- [ ] $= 1 \cdot 4 + 2 \cdot 3$
- [x] $= 1 \cdot 3 + 2 \cdot 4$

  Nice work

- [ ] $= \begin{pmatrix} 1 \times 3 \\ 2 \times 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 8 \end{pmatrix}$

3. Compute $\begin{pmatrix} -3 & 4 \\ 3 & -1 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ -5 \end{pmatrix}$. 1 / 1 point

Select all that apply.

- [x] $= \begin{pmatrix} -38 \\ 23 \end{pmatrix}$

  Nice work

- [ ] $= \begin{pmatrix} 38 \\ -23 \end{pmatrix}$
- [ ] none of the other options is correct
- [ ] $= \begin{pmatrix} (-3) \cdot 6 + 4 \cdot 6 \\ 3 \cdot (-5) + (-1)(-5) \end{pmatrix}$
- [ ] $= \begin{pmatrix} -38 & 23 \end{pmatrix}$
- [x] $= \begin{pmatrix} (-3) \cdot 6 + 4(-5) \\ 3 \cdot 6 + (-1)(-5) \end{pmatrix}$

  Nice work

4. Calculate the inverse of the matrix $A = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$. 1 / 1 point

Select all that apply.

- [x] The inverse is $\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}^{-1}$

  Nice work

- [ ] None of the other options is correct.
- [x] $A^{-1} = \frac{1}{3(2)-1(4)} \begin{pmatrix} 2 & -1 \\ -4 & 3 \end{pmatrix}$

  Nice work

- [x] The inverse is $\begin{pmatrix} 1 & -\frac{1}{2} \\ -2 & \frac{3}{2} \end{pmatrix}$

  Nice work

- [x] the inverse is $\frac{1}{2} \begin{pmatrix} 2 & -1 \\ -4 & 3 \end{pmatrix}$

  Nice work

- [x] $\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 & -\frac{1}{2} \\ -2 & \frac{3}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

  Nice work

  Yes, the matrix A multiplied by its inverse is the Identity Matrix . Well done

- [ ] $\det \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} = 0$

5. Using the system of linear equations and inverse matrix, solve the following simultaneous equations: 1 / 1 point

$$3x + y = 5$$
$$2x - y = 0$$

Select all the correct statements

- [x] our equations could be written as

  $$\begin{pmatrix} 3 & 1 \\ 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$$

  Nice work

- [x] We find the inverse of the coefficient matrix

  $$= \frac{1}{3(-1)-1(2)} \begin{pmatrix} -1 & -1 \\ -2 & 3 \end{pmatrix} = -\frac{1}{5} \begin{pmatrix} -1 & -1 \\ -2 & 3 \end{pmatrix}$$

  Nice work

- [x] The next step is to multiply both sides of our matrix equation by the inverse matrix:

  $$-\frac{1}{5} \begin{pmatrix} -1 & -1 \\ -2 & 3 \end{pmatrix} \cdot \begin{pmatrix} 3 & 1 \\ 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix}$$

  Nice work

- [x] $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

  Nice work

- [ ] None of the other options is correct